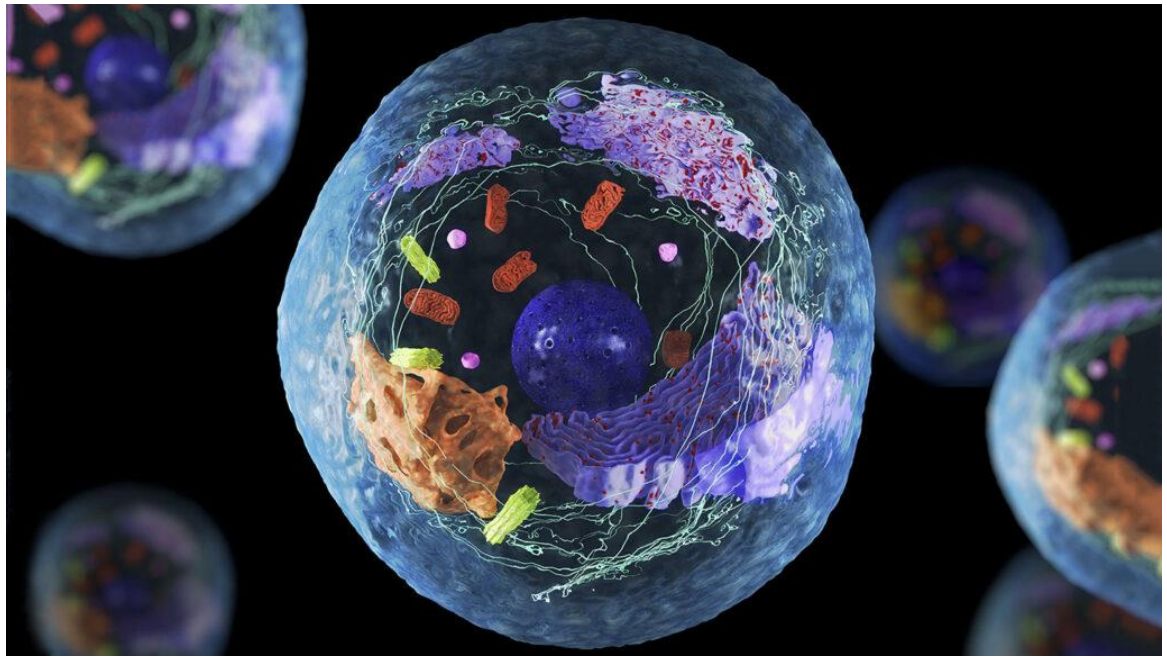
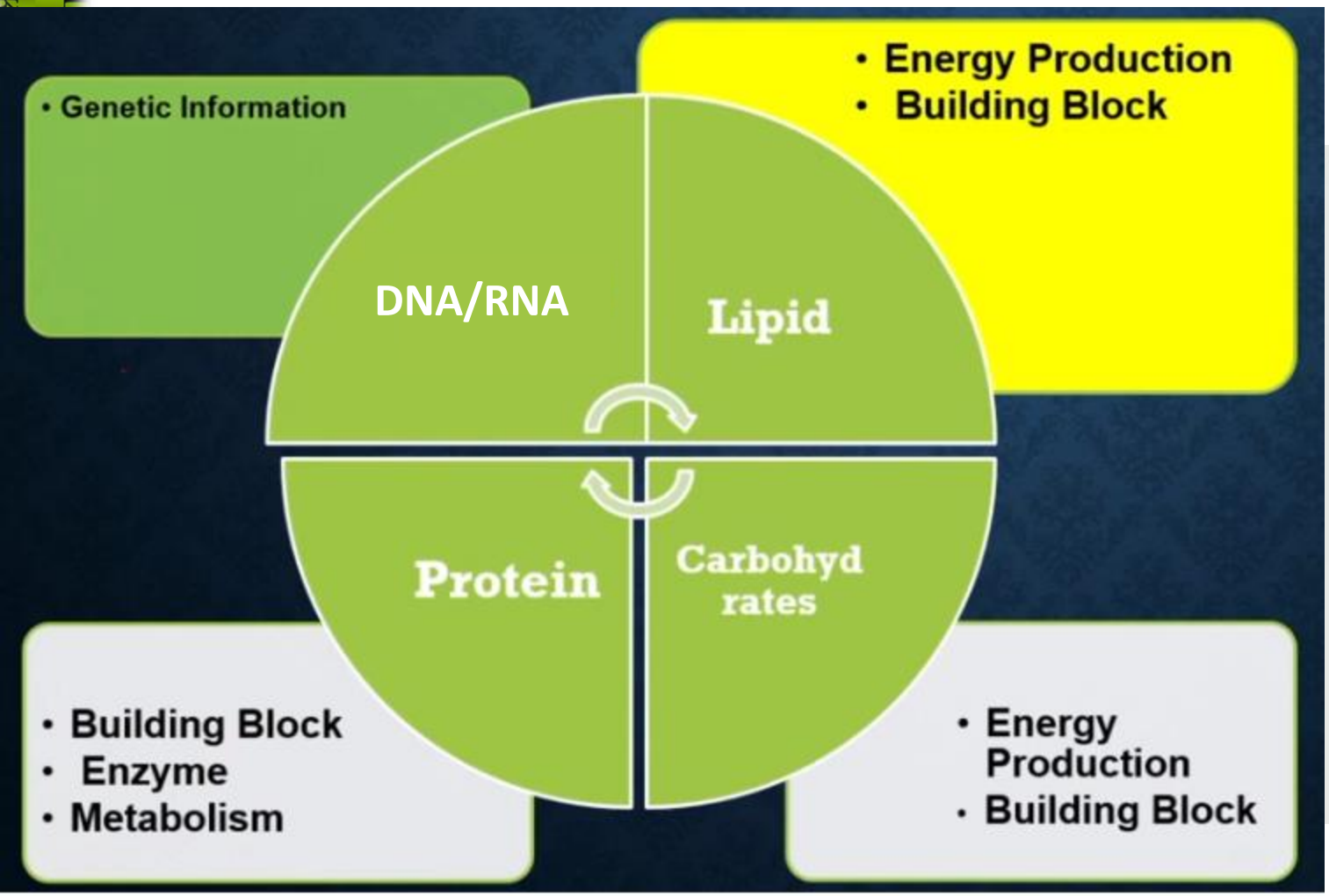
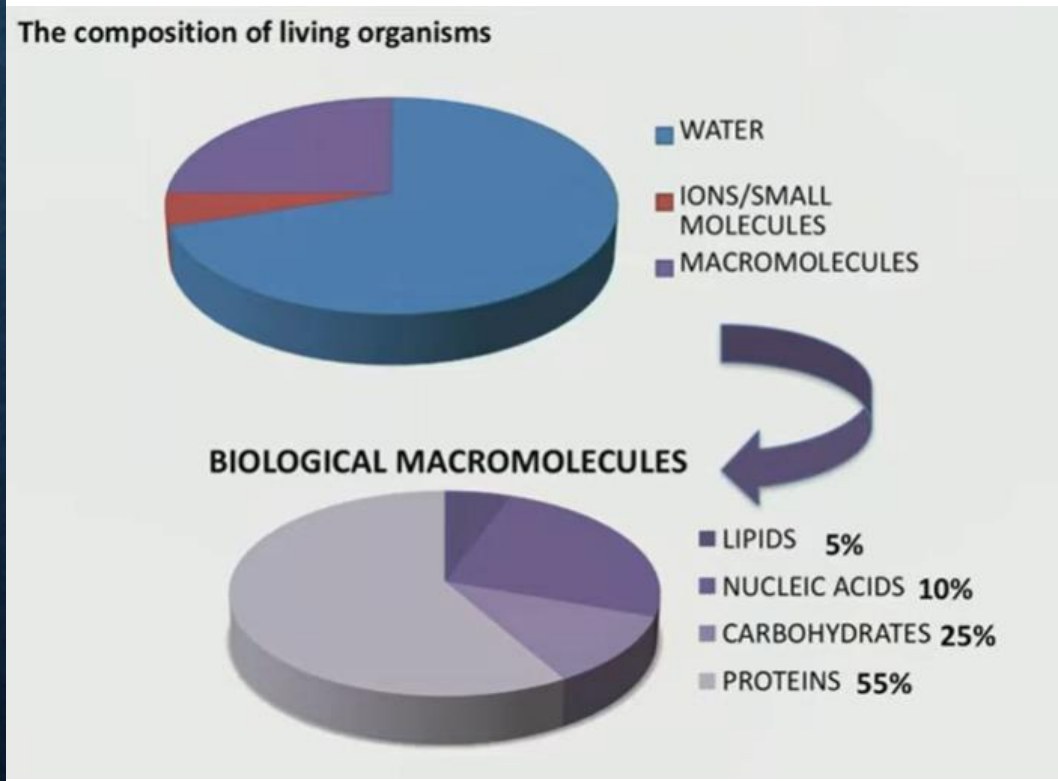
Lipids and Cell Membrane

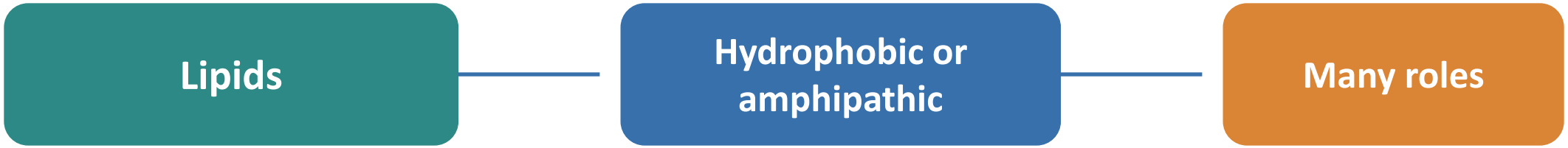
Lecture 13




• Genetic Information
• Energy Production
• Building Block
DNA/RNA
Lipid
Protein
Carbohyd rates
• Building Block
• Enzyme
• Metabolism
• Energy Production
• Building Block


The composition of living organisms
WATER
IONS/SMALL MOLECULES
MACROMOLECULES
BIOLOGICAL MACROMOLECULES
LIPIDS 5%
NUCLEIC ACIDS 10%
CARBOHYDRATES 25%
PROTEINS 55%

What are lipids?



- Poorly soluble in water is a common feature, but not a strict definition.
- Some lipids have both hydrophobic and hydrophilic regions.
- Lipids include chemically diverse molecules.

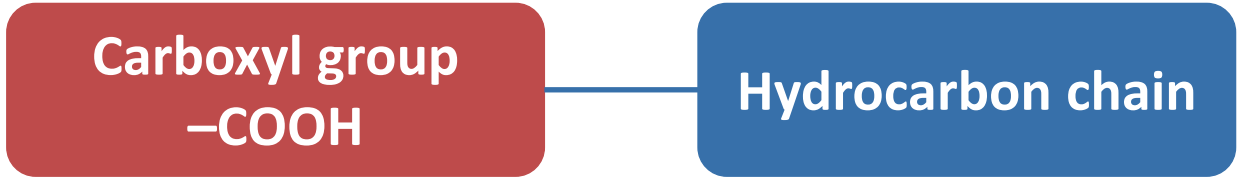
Major functions of lipids

Energy storage	Triacylglycerols store large amounts of chemical energy
Signaling	Some lipids act as hormones or signaling precursors
Membranes	Phospholipids and sterols create selective barriers
Protection	Insulation, cushioning and waterproofing

Lipid characterization

- Fats – Energy storage
- Phospholipids – Membrane structure
- Steroids – Hormonal
- Wax – Coating of animals and plants

Fatty acids: a key lipid building block



General form: $CH_3–(CH_2)n–COOH$

Hydrocarbon chain → mostly nonpolar

Carboxyl group → polar/ionizable

Chain length and double bonds strongly affect physical properties

Fatty acids: small structural changes, large effects

Saturated

no C=C bonds

straight chains • tight packing

cis-unsaturated

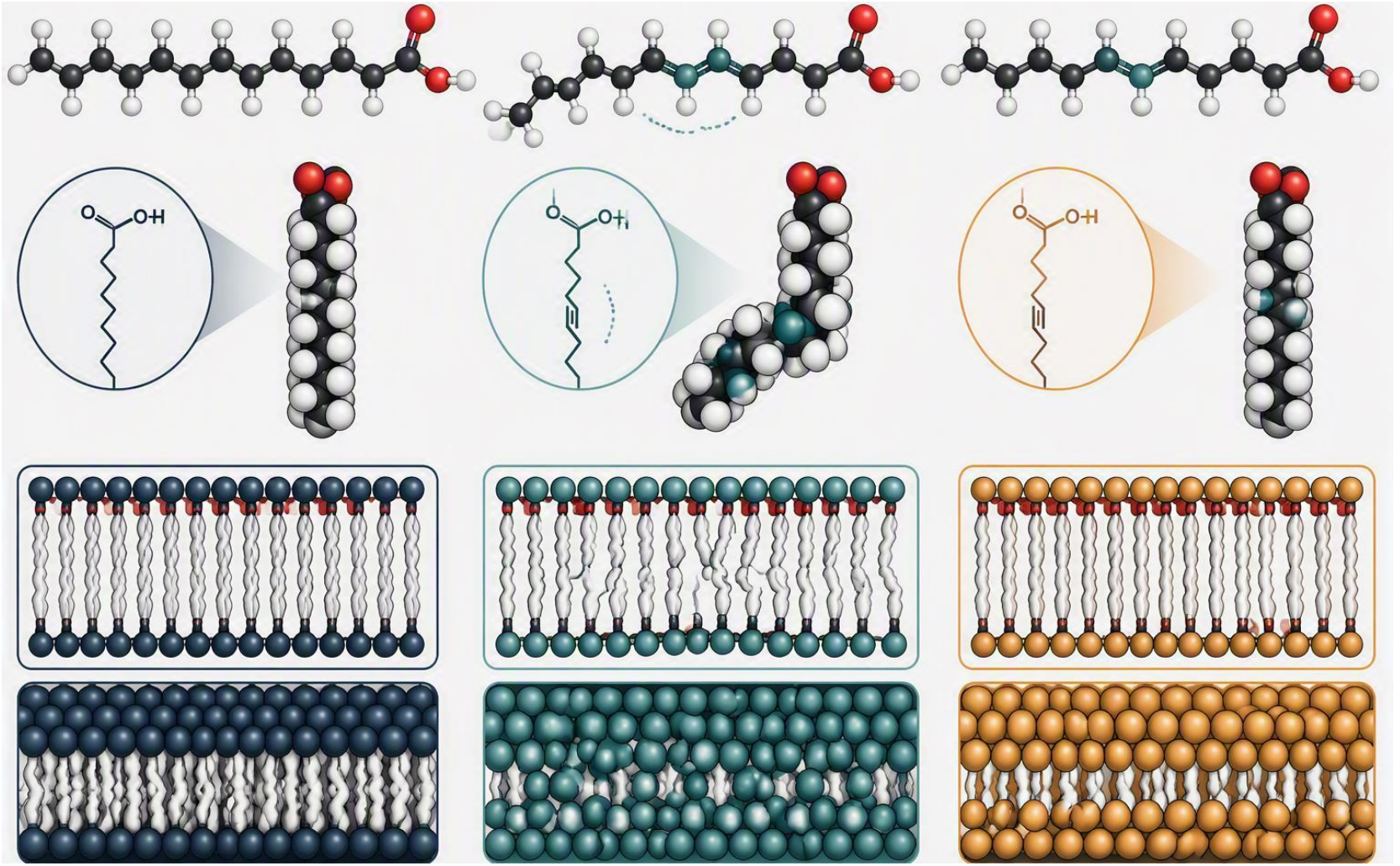
cis C=C bond

kinked chains • loose packing

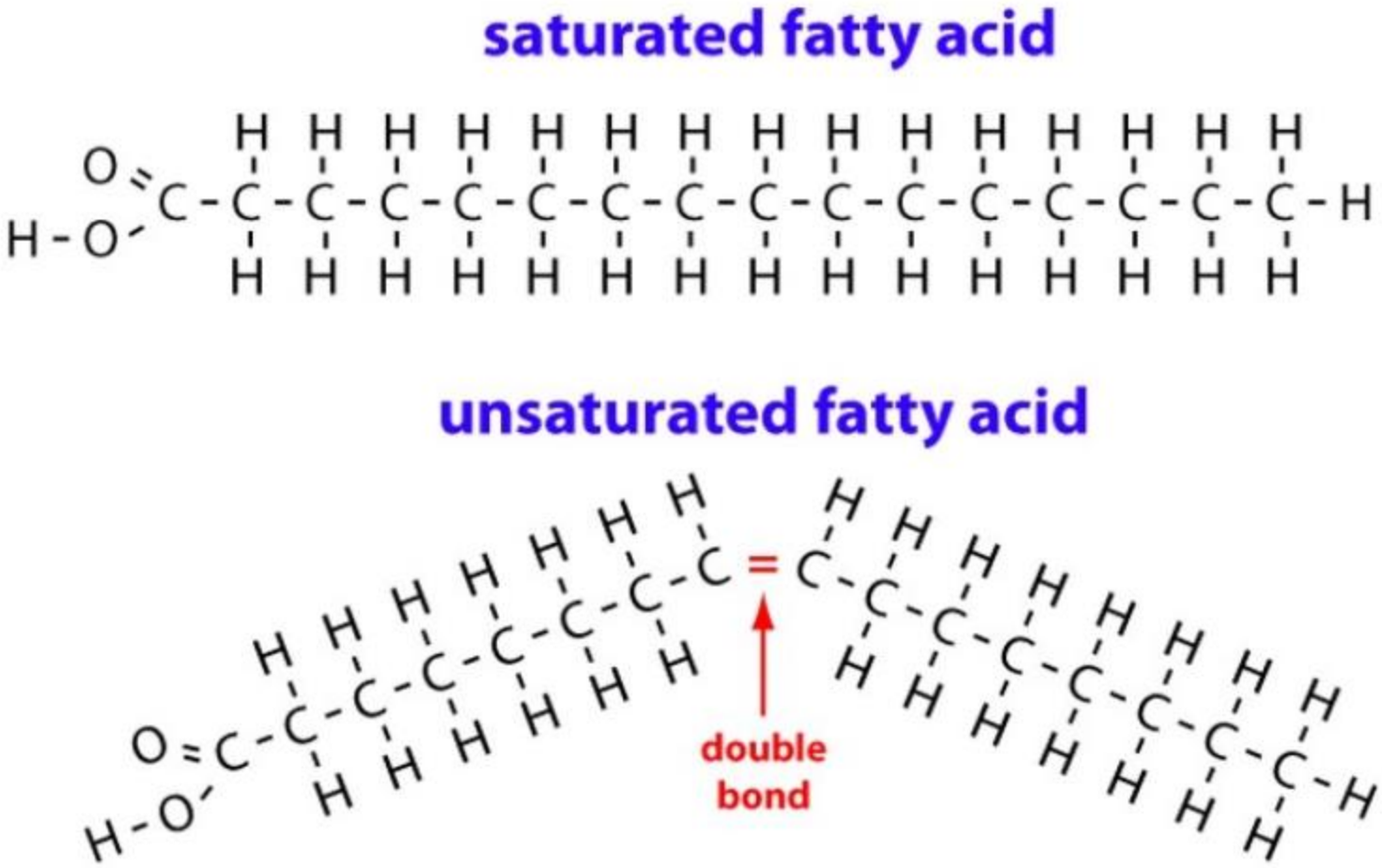
trans-unsaturated

trans C=C bond

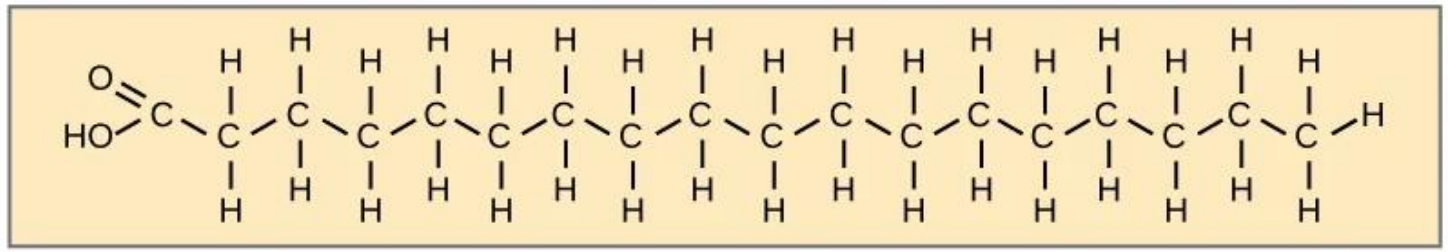
near-linear • tighter packing



Fatty acid

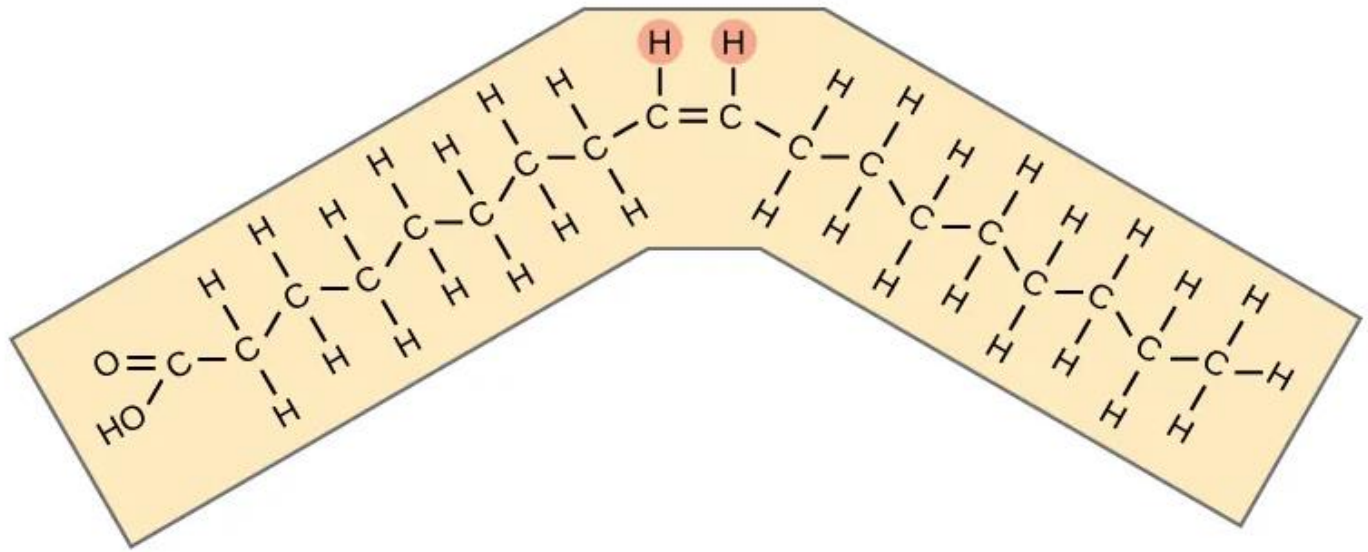


Stearic acid

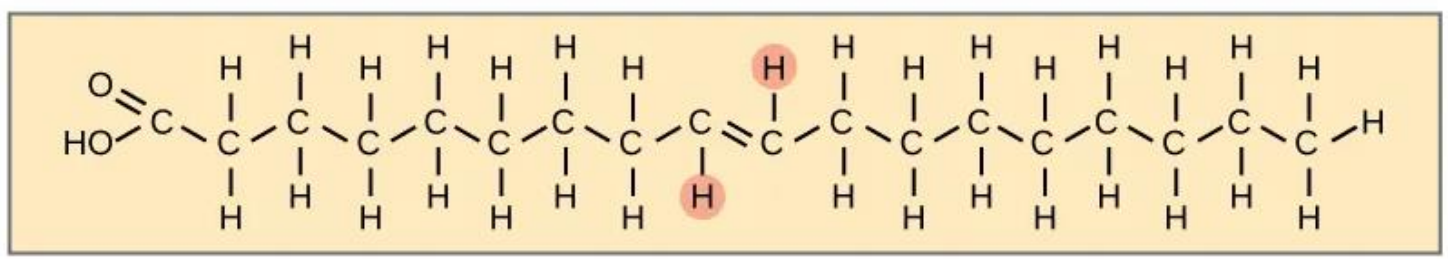


Unsaturated fatty acids

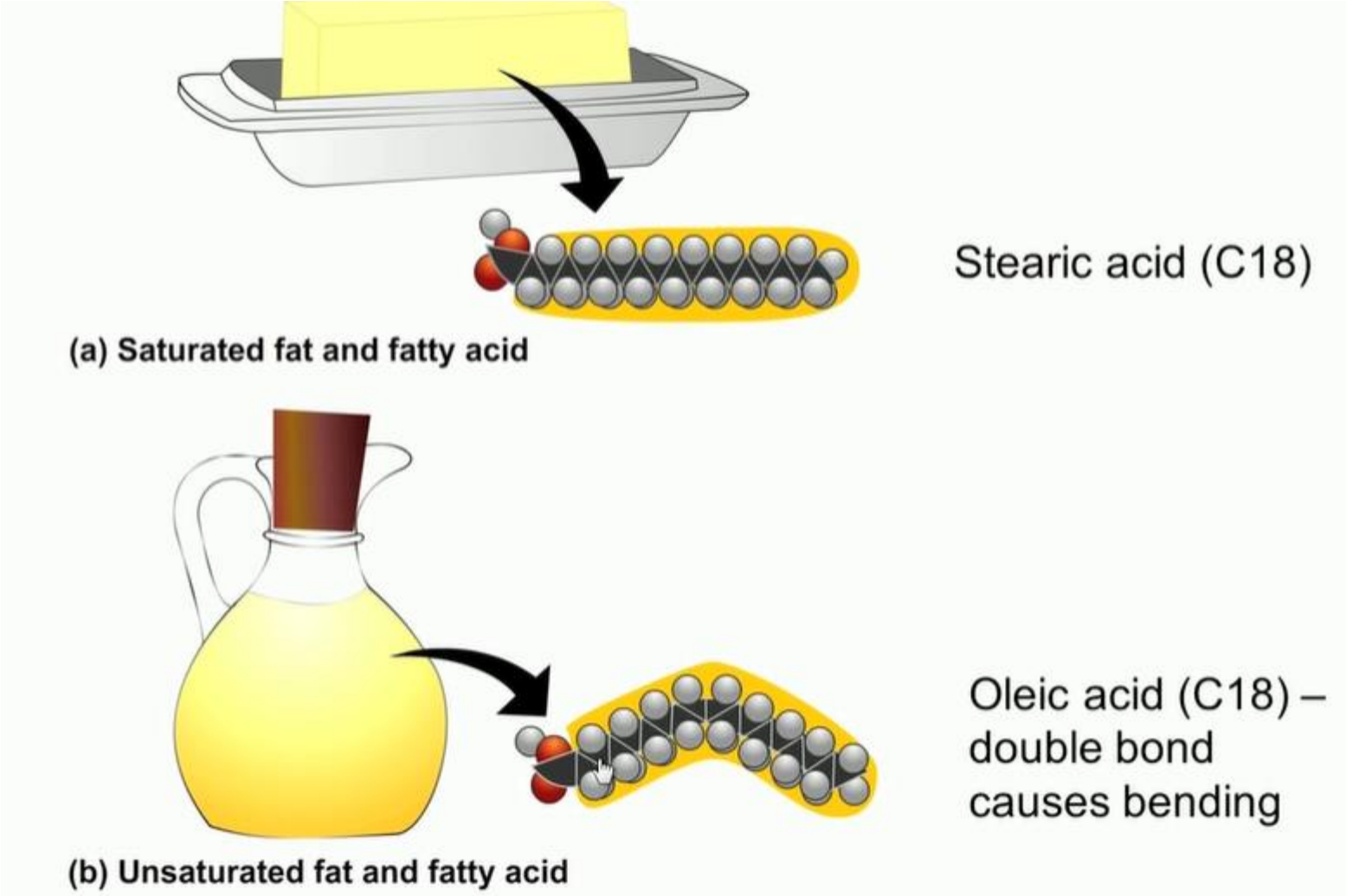
Cis oleic acid



Trans oleic acid



Fats


Stearic acid (C18)
(a) Saturated fat and fatty acid
Oleic acid (C18) – double bond causes bending
(b) Unsaturated fat and fatty acid

Triacylglycerols: compact energy storage

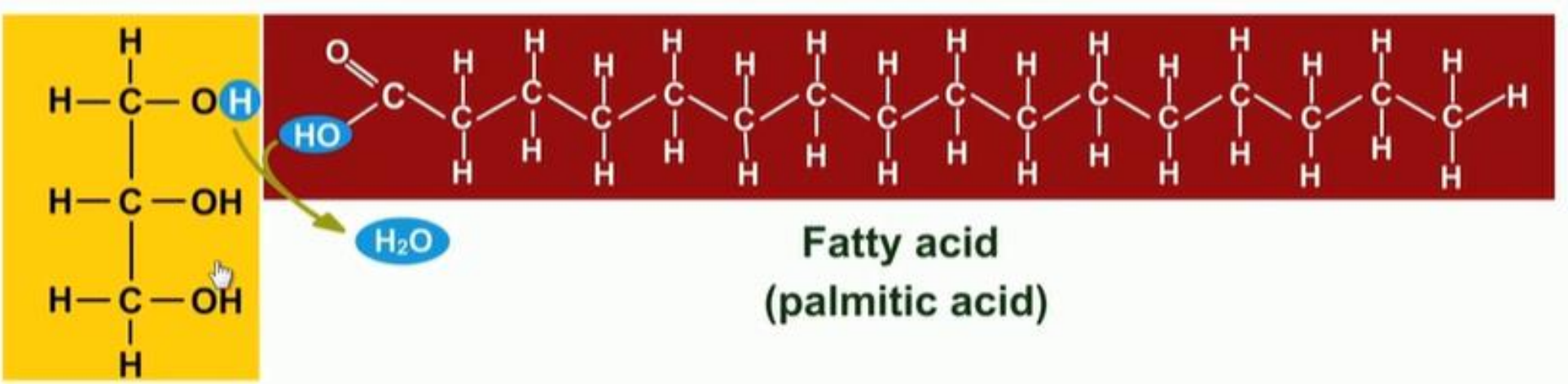


Ester bonds connect glycerol to three fatty-acid chains.

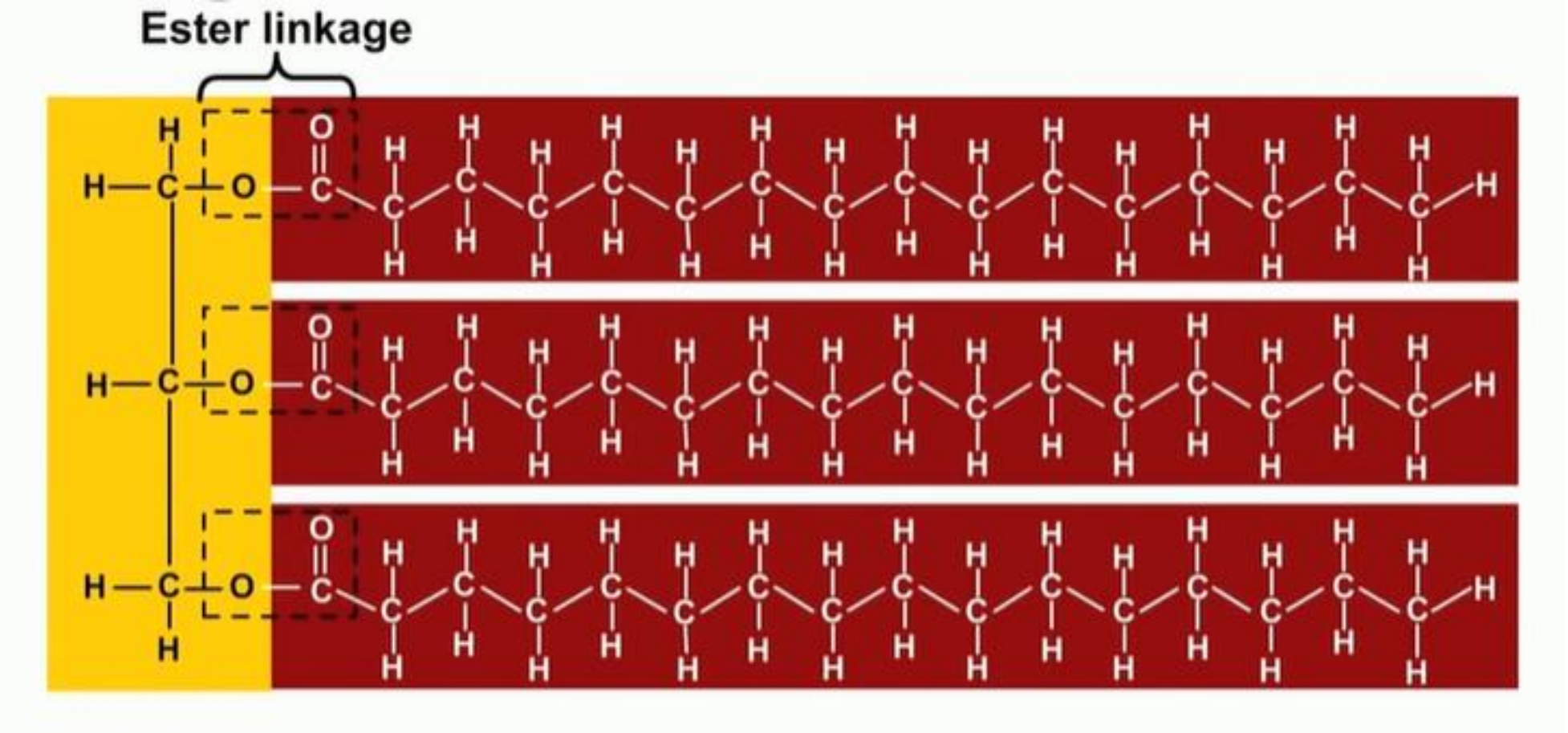
- Highly reduced carbon stores substantial chemical energy.
- Hydrophobic molecules can be stored with little associated water.
- Major long-term energy reserve in animals.

Glycerol + Fatty acid → Fat (Triacylglycerols)

Lipids are fatty, waxy, or oily compounds that are soluble in organic solvents and insoluble in polar solvents such as water.



Lipids



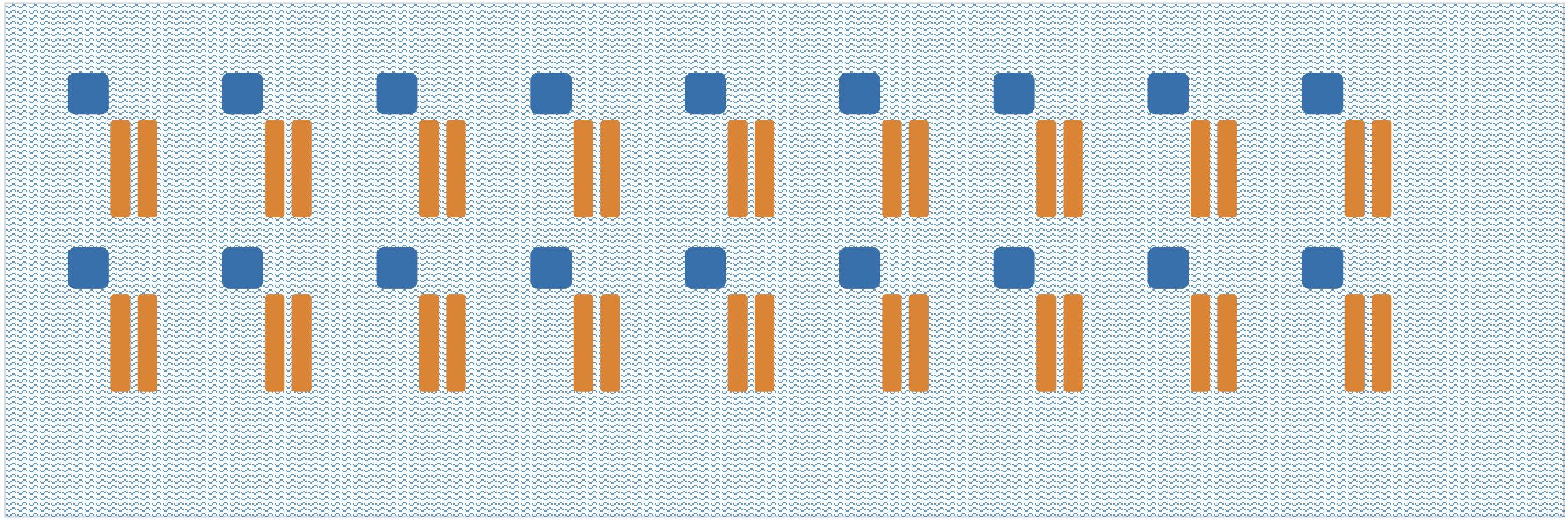
Triglyceride (Fat)

Phospholipids: amphipathic molecules



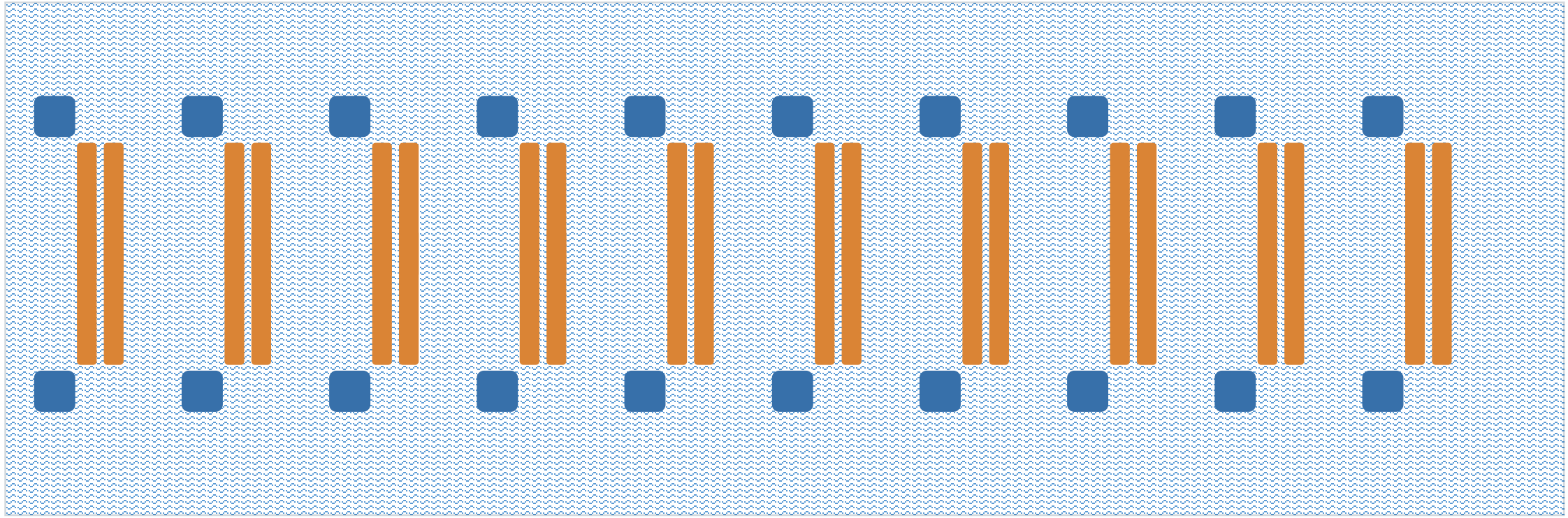
Amphipathic = contains both **hydrophilic** and **hydrophobic** regions

Why do phospholipids form bilayers?



Hydrophobic tails avoid water; polar heads interact with water.

The lipid bilayer is a selective barrier

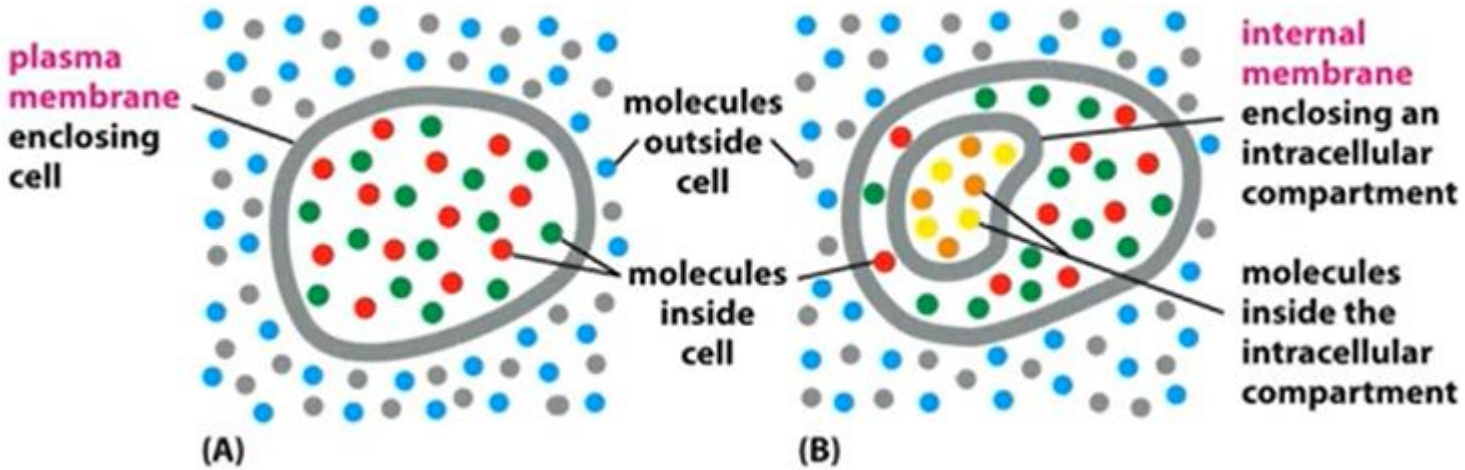


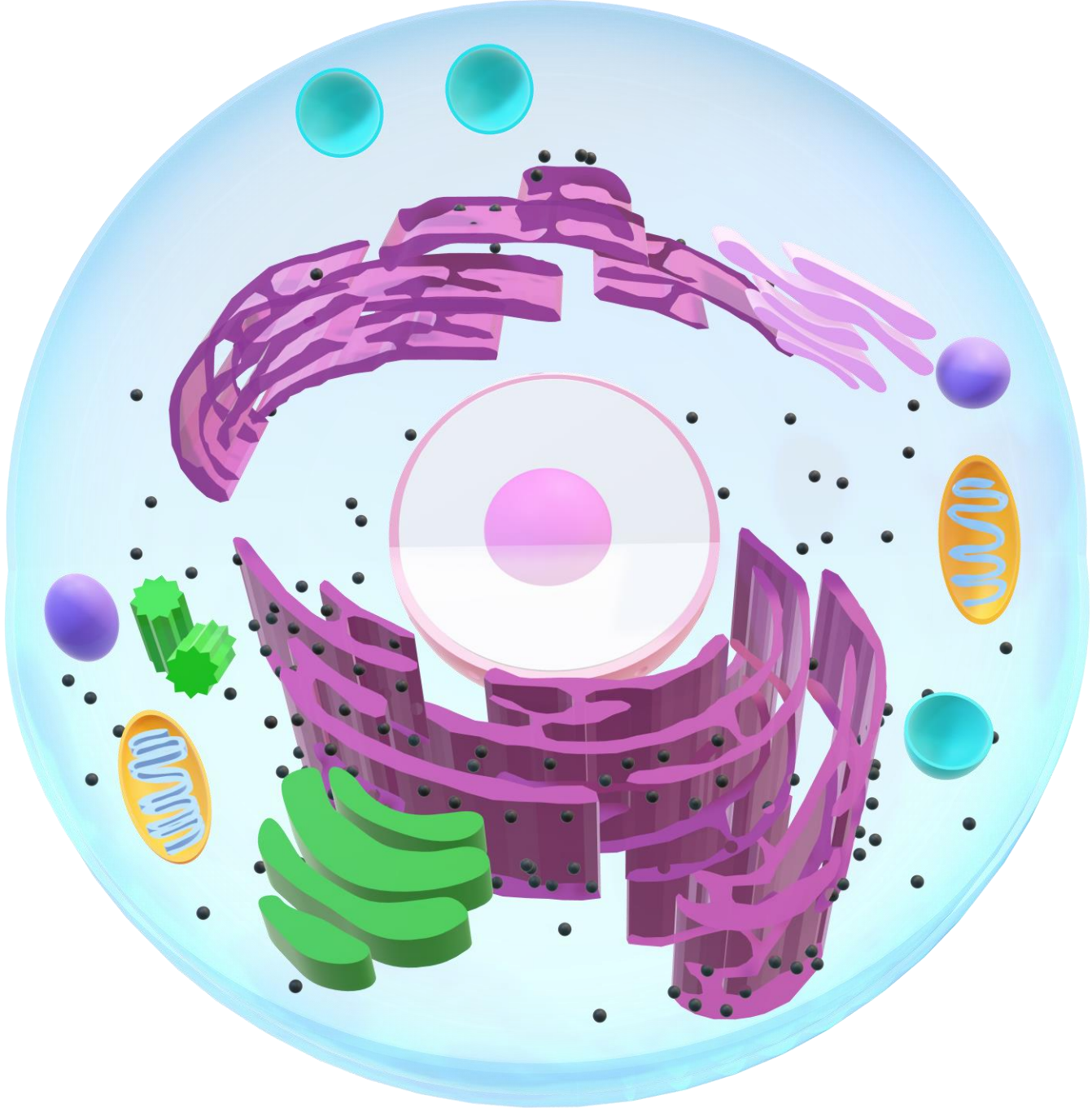
Small nonpolar molecules cross relatively easily; ions and many polar molecules need transport proteins.

Cell Membrane

- Every known cell has a cell membrane
  - Thin layer, made up of lipids.

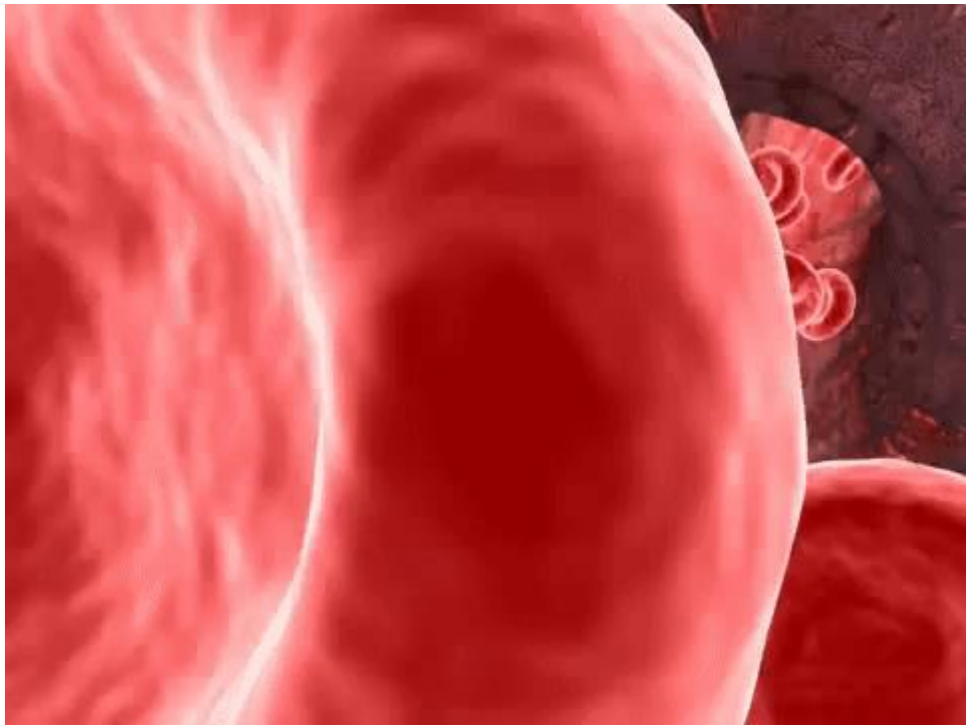
- Separates the cell from outside environment
  - partitions material inside the cell from outside
  - Maintains order inside the cell





https://sketchfab.com/

Importance of cell membrane

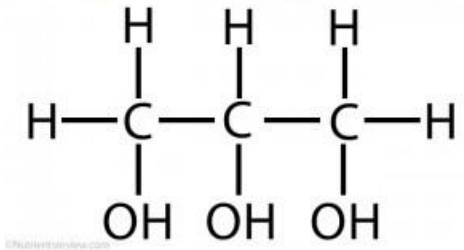


www.medflix.com

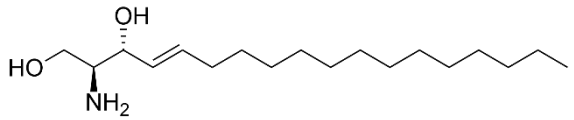


Structure of membrane lipids

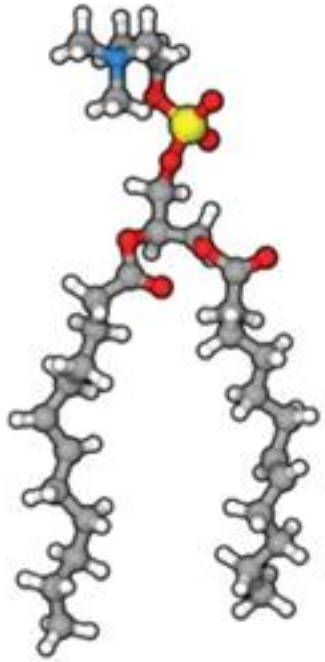
Glycerol (Glycerin)



Sphingosine

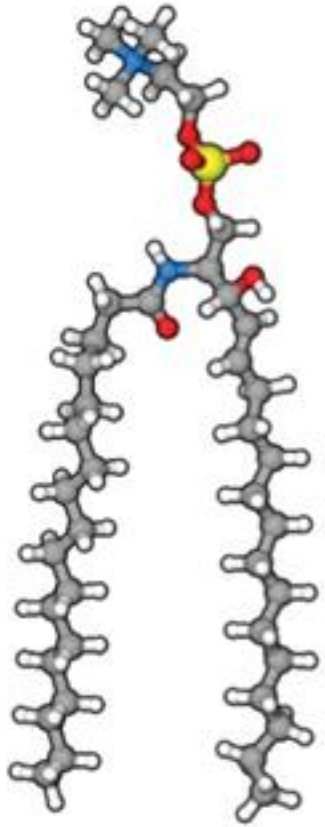


(A)



(glycerol-based) phospholipid

(B)



sphingolipid

(C)



cholesterol

Cholesterol: a membrane 'buffer'

- Rigid ring structure inserts among phospholipid tails.
- At moderate temperatures, it restrains excessive lipid movement.
- At low temperatures, it helps prevent tight packing.
- It also reduces permeability to some small water-soluble molecules.



Lipid membrane: general structure

simplified lipid structure

lipid bilayer

lipid bilayer vesicle

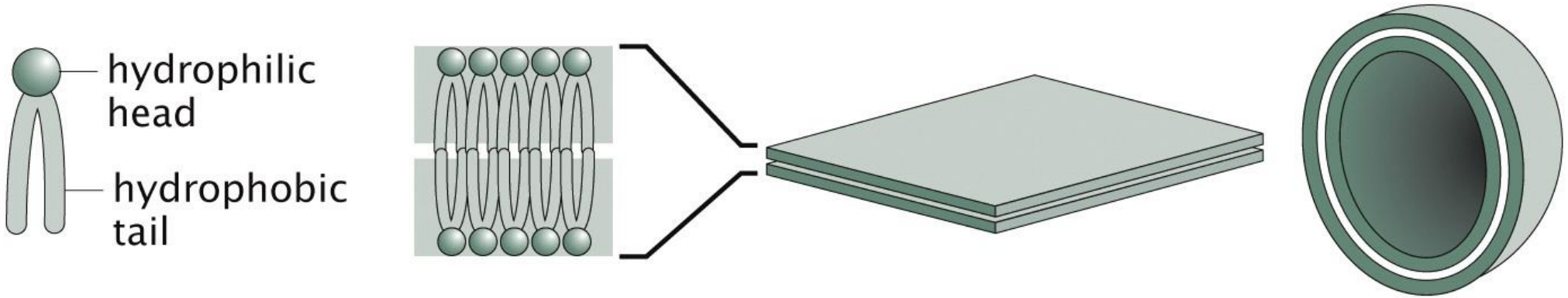
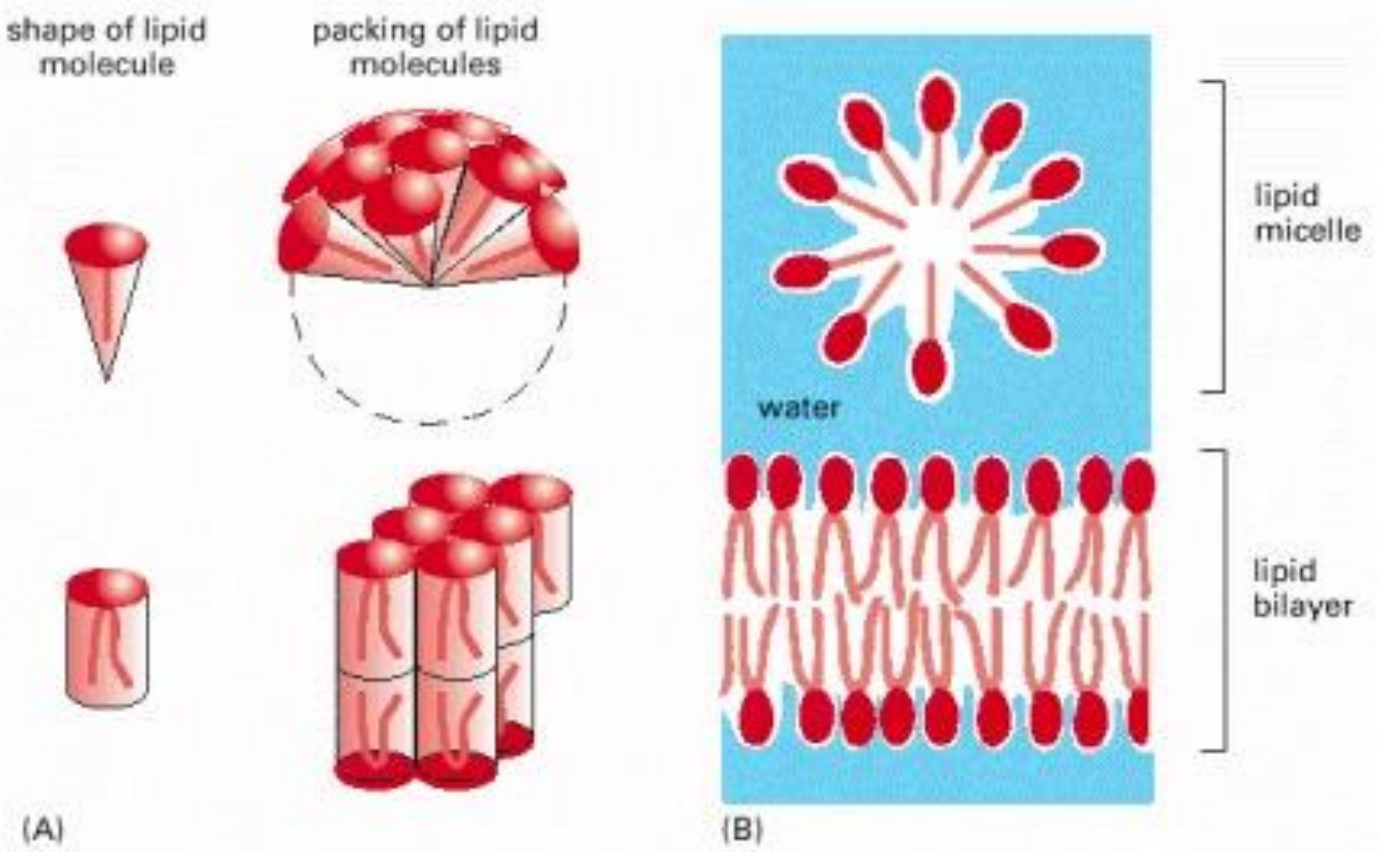


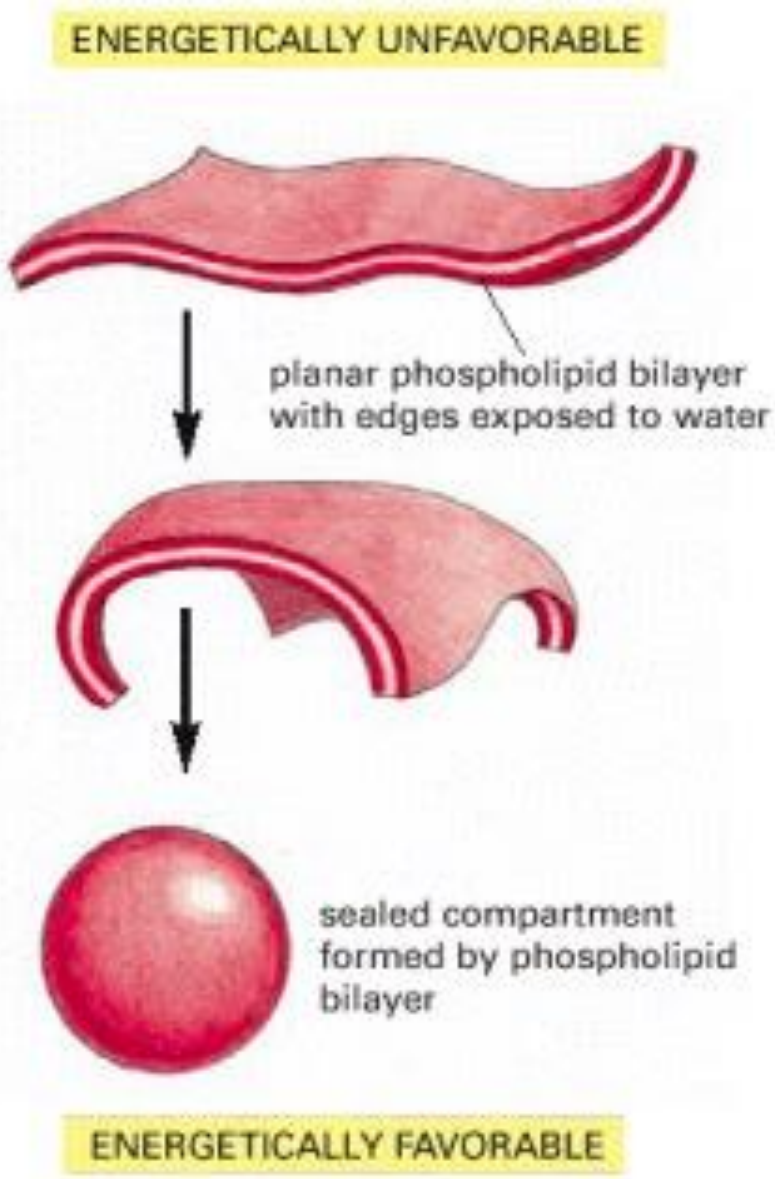
Figure 11.1a Physical Biology of the Cell, 2ed. (© Garland Science 2013)

avoidance of edge effects

Membranes are about 4-5 nm thick

Lipid membrane: general structure





https://www.ncbi.nlm.nih.gov/books/NBK26871/

Geometry of lipids determines membrane shape

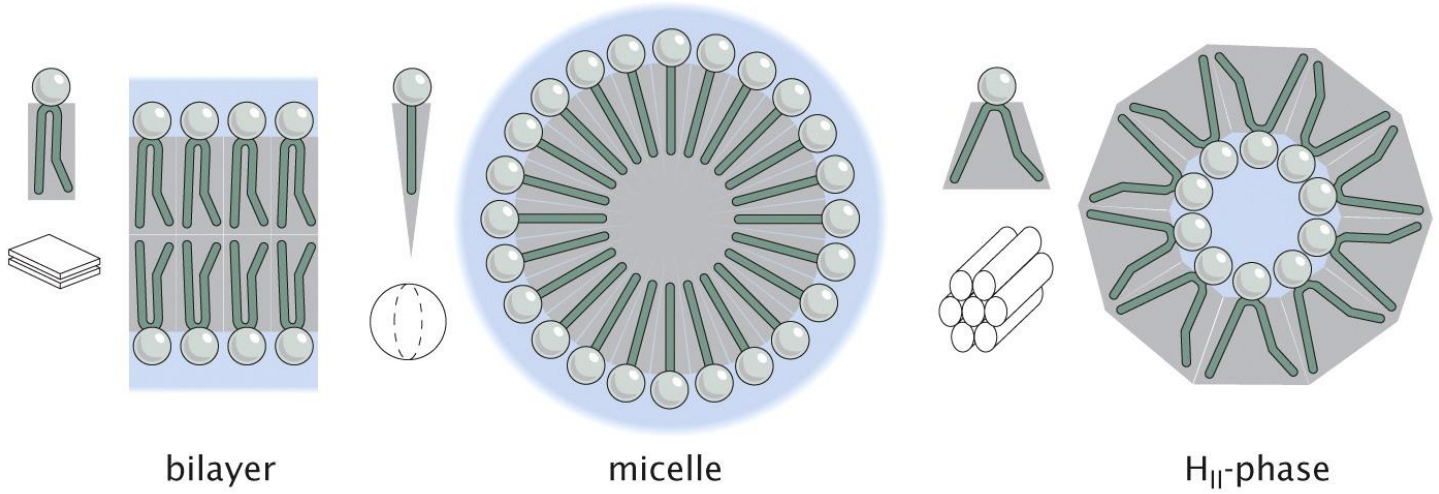


Figure 11.7 Physical Biology of the Cell, 2ed. (© Garland Science 2013)

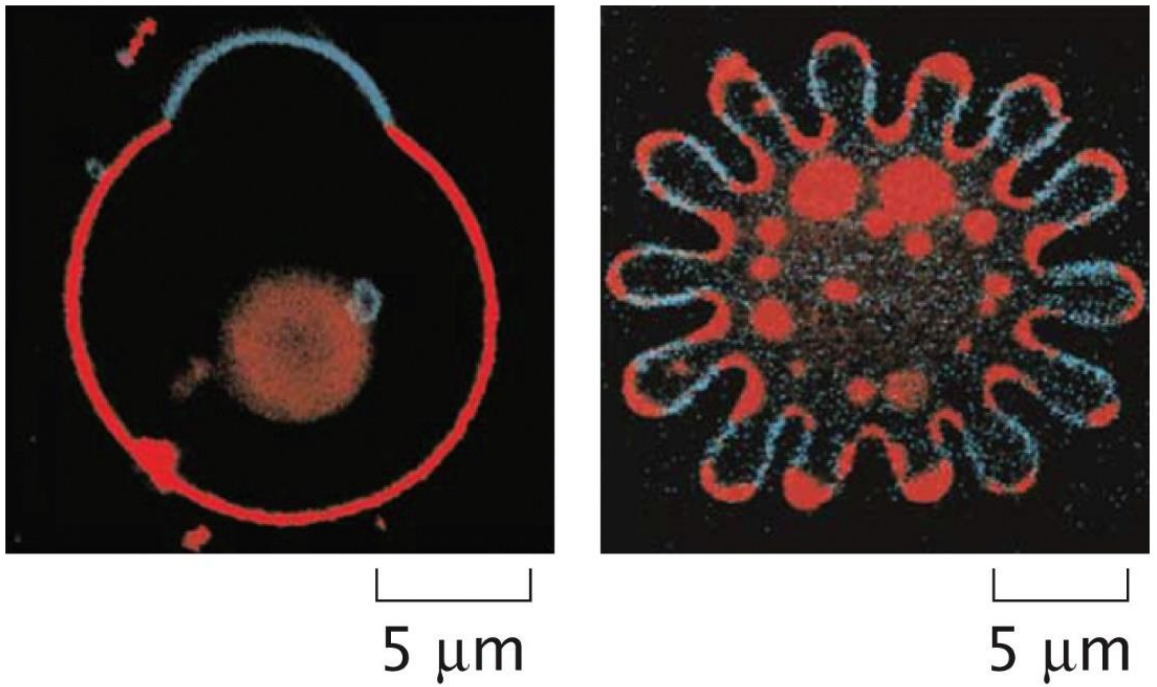
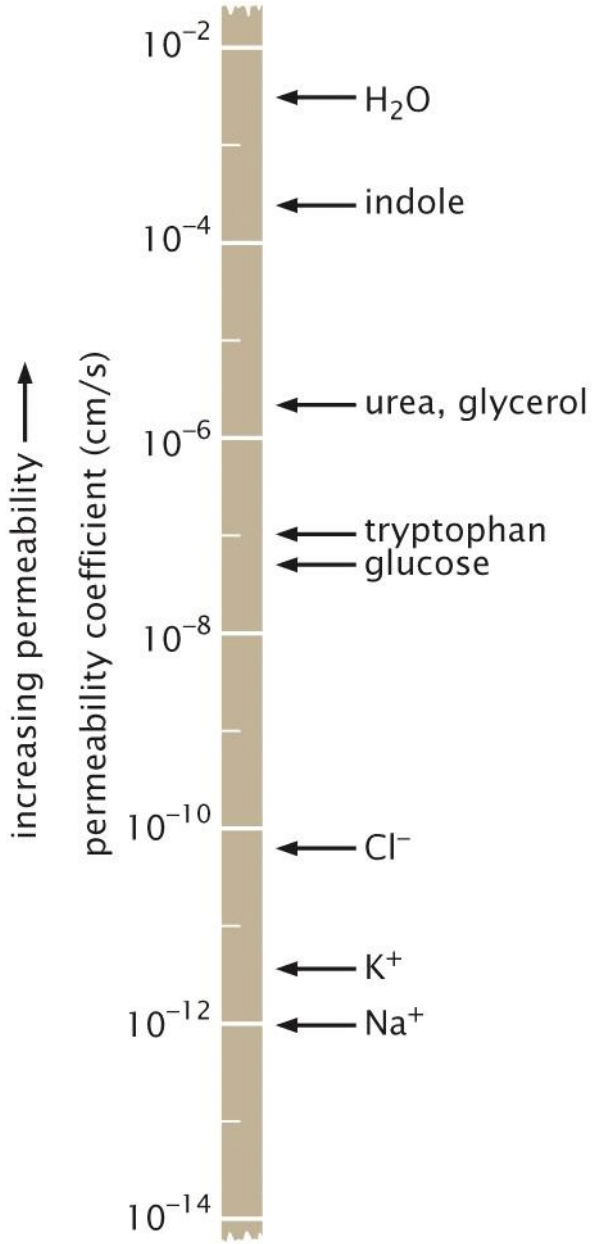


Figure 11.8 Physical Biology of the Cell, 2ed. (© Garland Science 2013)

Structure → property → function: the lipid framework

Hydrocarbon chain length	Packing / melting behavior
Double bonds	Shape / fluidity / oxidation
Head-group chemistry	Charge / interactions

Membrane transport



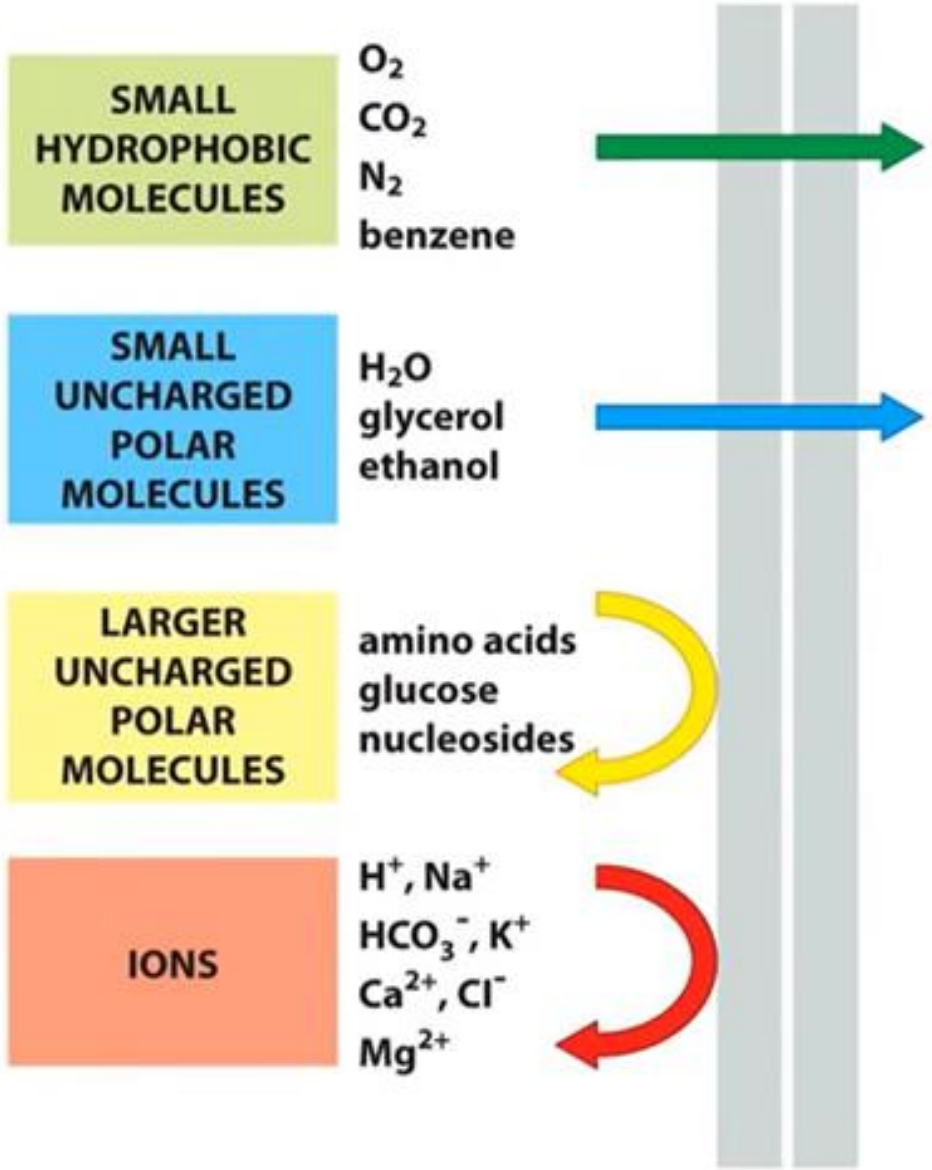
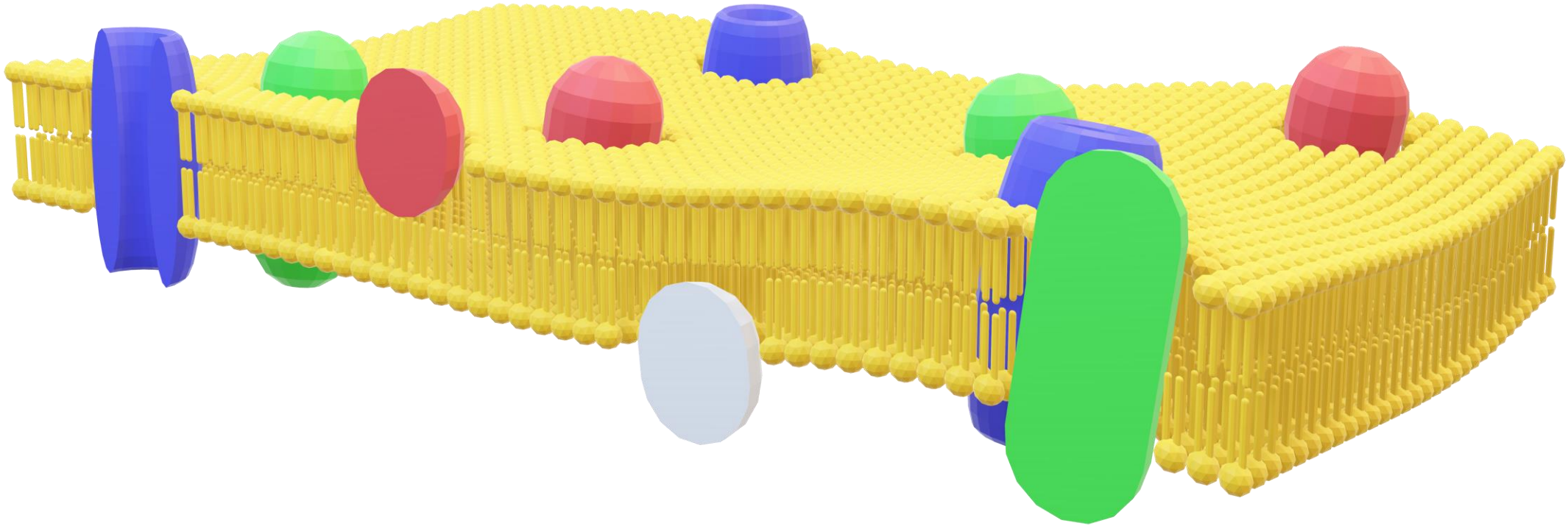


Figure 11.11 Physical Biology of the Cell, 2ed. (© Garland Science 2013)

Cell Membrane



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